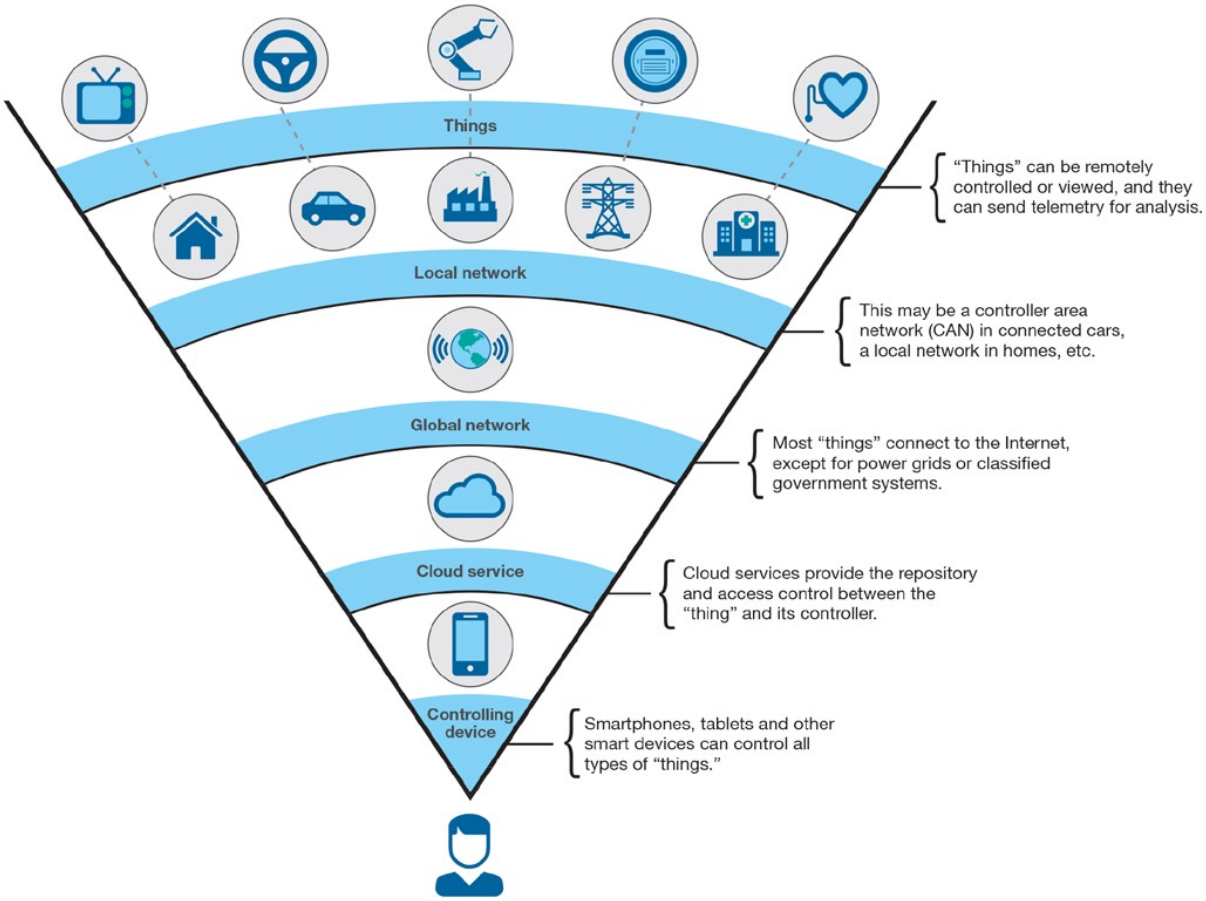
IoT Internet of Things

IoT Cloud provider point of view

Hartmut Seitter

IBMs view of - Simplify the development of your IoT solutions with IoT architectures



Components of an IoT Reference Architecture

The details of reference architectures differ depending on the application domain; however, most IoT reference architectures describe at least the following capabilities:

- **Managing devices and their data**
- **Connectivity and communication**
- **Analytics and applications**

Connectivity and communication

- **Managing connectivity and bidirectional communication between devices, between devices and gateways, and between gateways and cloud services and applications is another key capability that is often described within IoT reference architectures. For edge computing, event-driven architectures are a good choice, with communication between devices and services using publish/subscribe protocols and message brokers.**

Analytics and applications

- **To derive value from the data that comes from IoT devices, cloud applications provide visualization and analytics tools that operate over streams or batches of data to identify actionable insights. Depending on the use case, decision management and business process tools can trigger alerts or carry out actions in response.**

Making sense of IoT data – IBMs point of view

- **Storing Data**
  - Devices itself have limited storage capabilities and very often battery powered.
  - The data collected by the devices is communicated using protocols (MQTT, CoAP, HTTP, LoRaWAN, etc.) to central IoT services for processing.
- **Heterogeneous data**
  - Data captured by the devices are a mix of data formats. E.g. structured, unstructured, semi-structured. So, no one-size-fits-all-approach exits for storing IoT data.
- **Transforming data**
  - Be sure to transform your data and then be selective about which data to transmit and store to save on bandwidth and storage costs.
- **Securing data**
  - Data must be transmitted and stored securely in order to maintain data integrity and privacy, using secure protocols and encryption.
- **Data Storage Technologies**
  - The important considerations in deciding the right data-storage strategy are volume of data, the connectivity to the network, and power availability.
  - E.g. Autonomous cars can generate Petabytes of data in a year, does it make sense to transfer every bit?
  - E.g. Aircrafts may have intermediate connectivity

  - NoSQL databases are a popular choice for IoT data
  - Time series databases can be relational or based on a NoSQL approach.

  - Apache Hadoop is an open source framework to store and process big data.

Making sense of IoT data – IBMs point of view

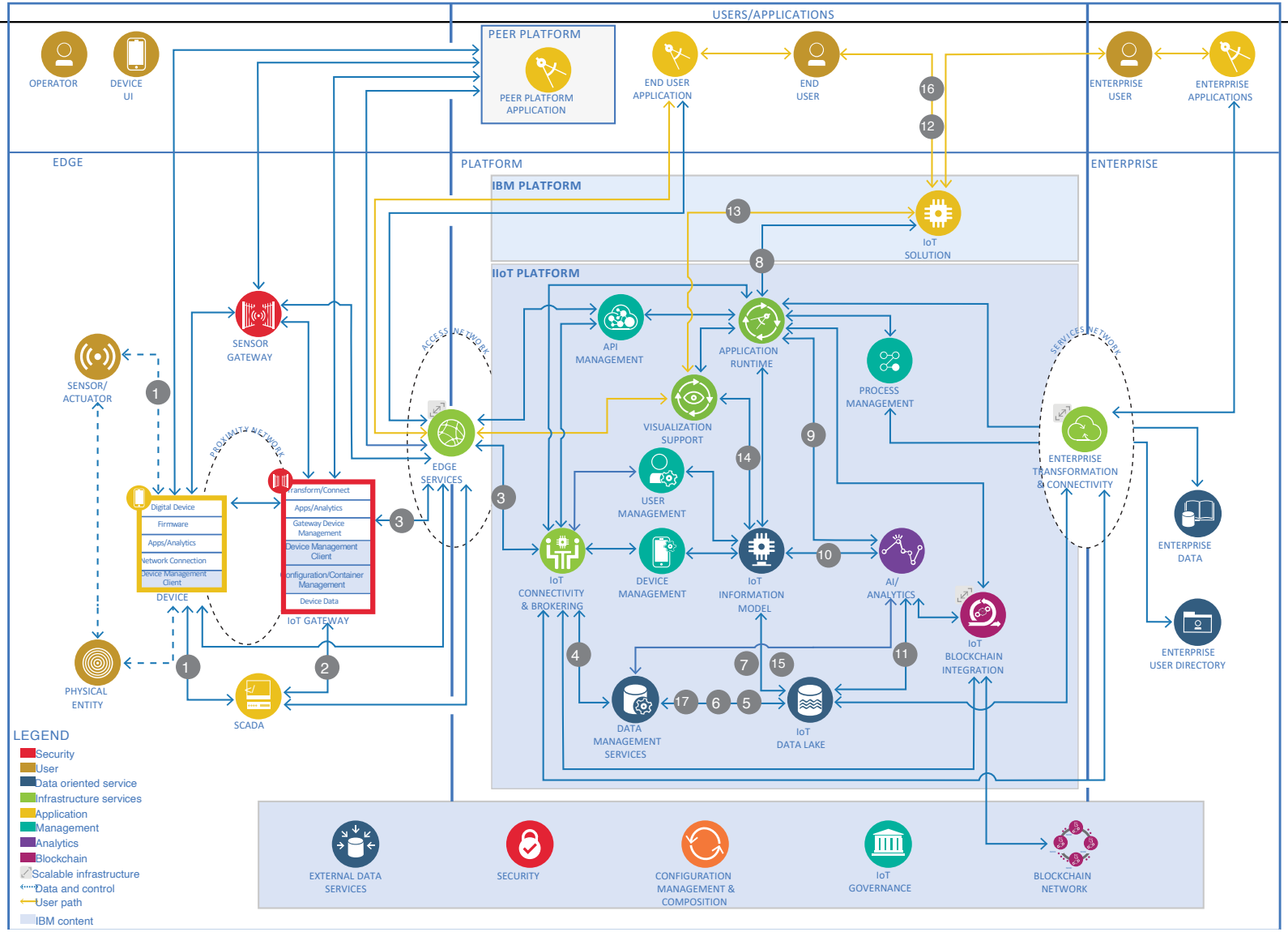
- **Analyzing data**

  IoT data needs to be analyzed in order to make it useful – remember from ‘Data to Information’

  - **Distributed analytics**
    - Data can be spread across multiple database.
    - Analytics may involve aggregating results which are distributed across multiple databases or datastores and even across multiple geographical locations.
  - **(Near) Real-time analytics**
    - Real-time analytics are also ideal for time series data, because unlike batch processing, real-time analytics tools usually support controlling the window of time analysis, and calculating rolling metrics, for example, to track hourly averages over time rather than calculating a single average across an entire dataset.
  - **Edge Analytics**
    - The data is pre-processed to filter out duplicates or to re-order, aggregate or normalize the data prior to analysis
  - **Machine learning**
    - Machine learning can be applied to historic or real-time data. Machine learning techniques can be used to identify patterns, identify key variables and relationships between them to automatically create and refine analytics models, and then use those model for simulations or to produce decisions.

IoT Reference Architecture – IBM point of view - based on 2020!

Hartmut Seitter



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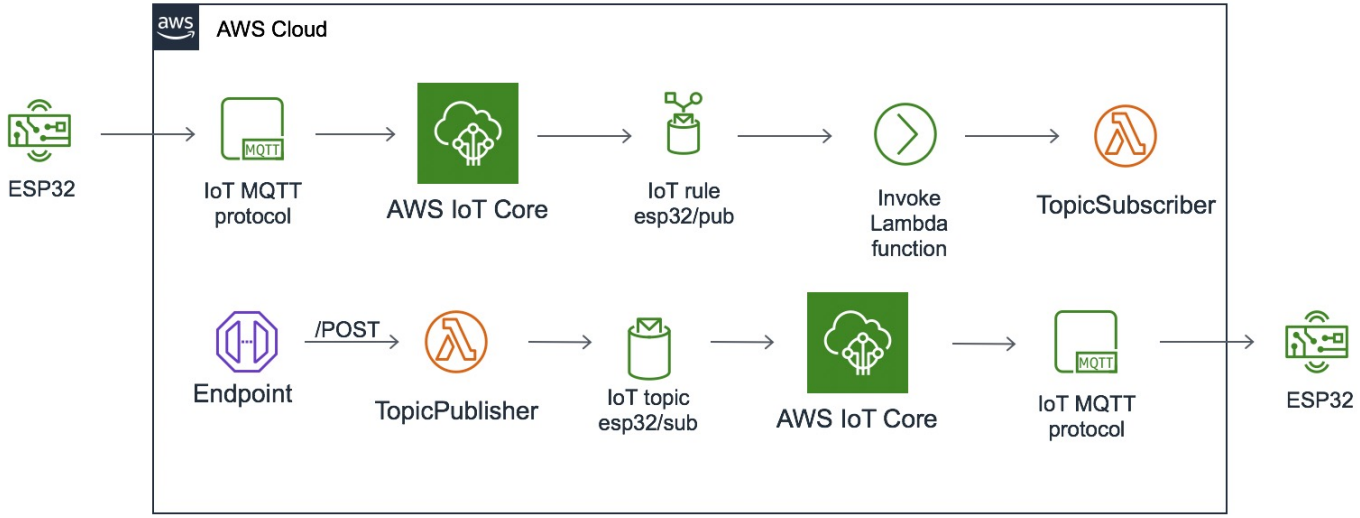
AWS Cloud Platform >>>>>>>> re:Invent

Re:Invent means they removed the IoT Reference Architecture Slides (from 2021) and description

Now they put all into AWS IoT Core



https://aws.amazon.com/blogs/compute/building-an-aws-iot-core-device-using-aws-serverless-and-an-esp32/

AWS IoT Core

Secure device connectivity: Enables secure, bi-directional communication over protocols like MQTT, HTTPS, and WebSockets. It also supports LoRaWAN for long-range, low-power devices.

Authentication and authorization: Includes built-in features for authenticating devices and managing access policies to safeguard your solution.

Message routing and processing: Can process and route messages from devices to other AWS services like Kinesis or SageMaker for further analytics, or to other devices.

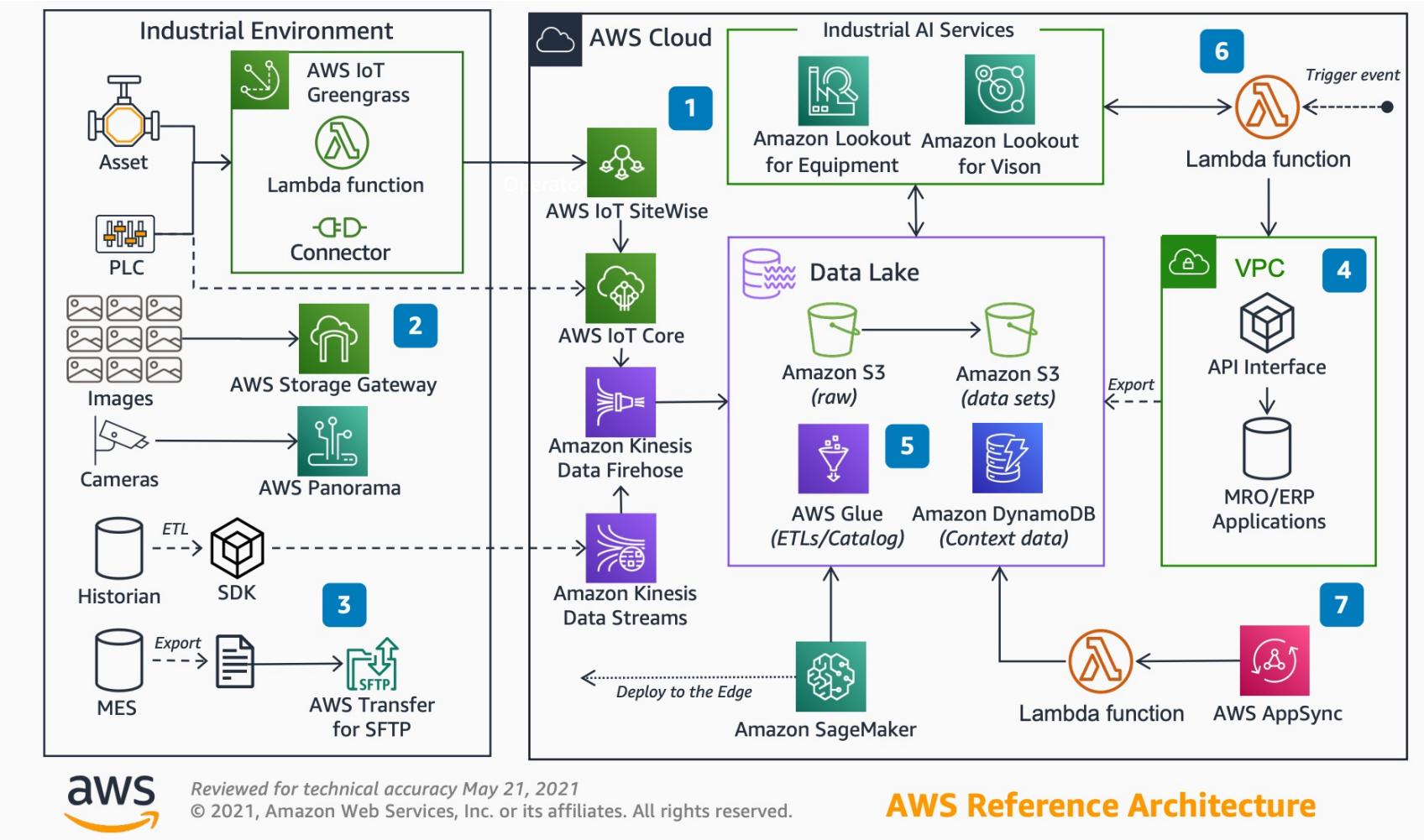
Device management: Facilitates device registration, management, and communication, even at a massive scale.

Application integration: Acts as a bridge between your devices and your cloud applications, allowing apps to access device data and send commands back to the devices.

Device testing: Offers tools like AWS IoT Core Device Advisor to test device functionality before onboarding them to the cloud.

https://aws.amazon.com/iot-core/features/

AWS Reference Architecture – Industrial Platform based on 2021



AWS Reference Architecture – Industrial Platform

1. Asset, machine, and PLC data is ingested and transformed using AWS IoT Greengrass connectors and AWS Lambda functions running on AWS IoT Greengrass at the Edge. Industrial Internet of Things (IIoT) data can then be streamed into the data lake from AWS IoT SiteWise via AWS IoT Core and Amazon Kinesis Data Firehose.

2. Images can be synchronized to the data lake using AWS Storage Gateway. For IP cameras, AWS Panorama can be used to transfer the images into the data lake.

3. Manufacturing application data can be streamed into the data lake using Amazon Kinesis Data Streams or exported and transferred using AWS Transfer for SFTP.

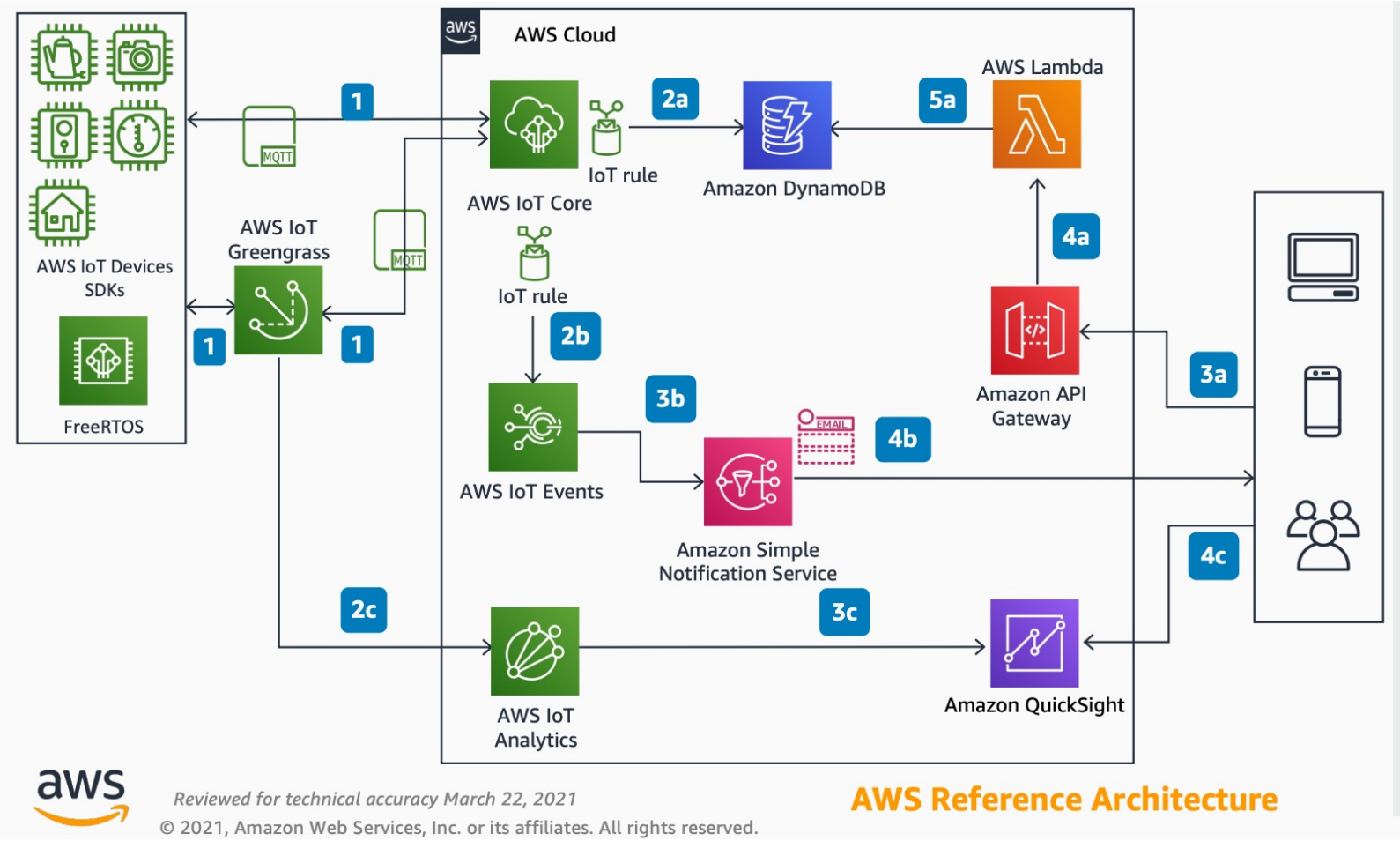
4. Industrial enterprise application data can be exported to the data lake using the application interfaces. Historical records can be stored in Amazon S3 and reference data can be stored in Amazon DynamoDB.

5. AWS Glue can be used for data engineering and cataloging data sets in the data lake.

6. Machine Learning (ML) models can be built using Amazon SageMaker, or AWS for Industrial AI services can be used to detect anomalies. The ML Inference can be initiated from a AWS Lambda function and results can be sent to the enterprise application using the applications API interface.

7. Data sets and results can be accessed by dashboards and applications using AWS AppSync.

Connected Home - Telemetry Measuring and collecting data from smart home devices



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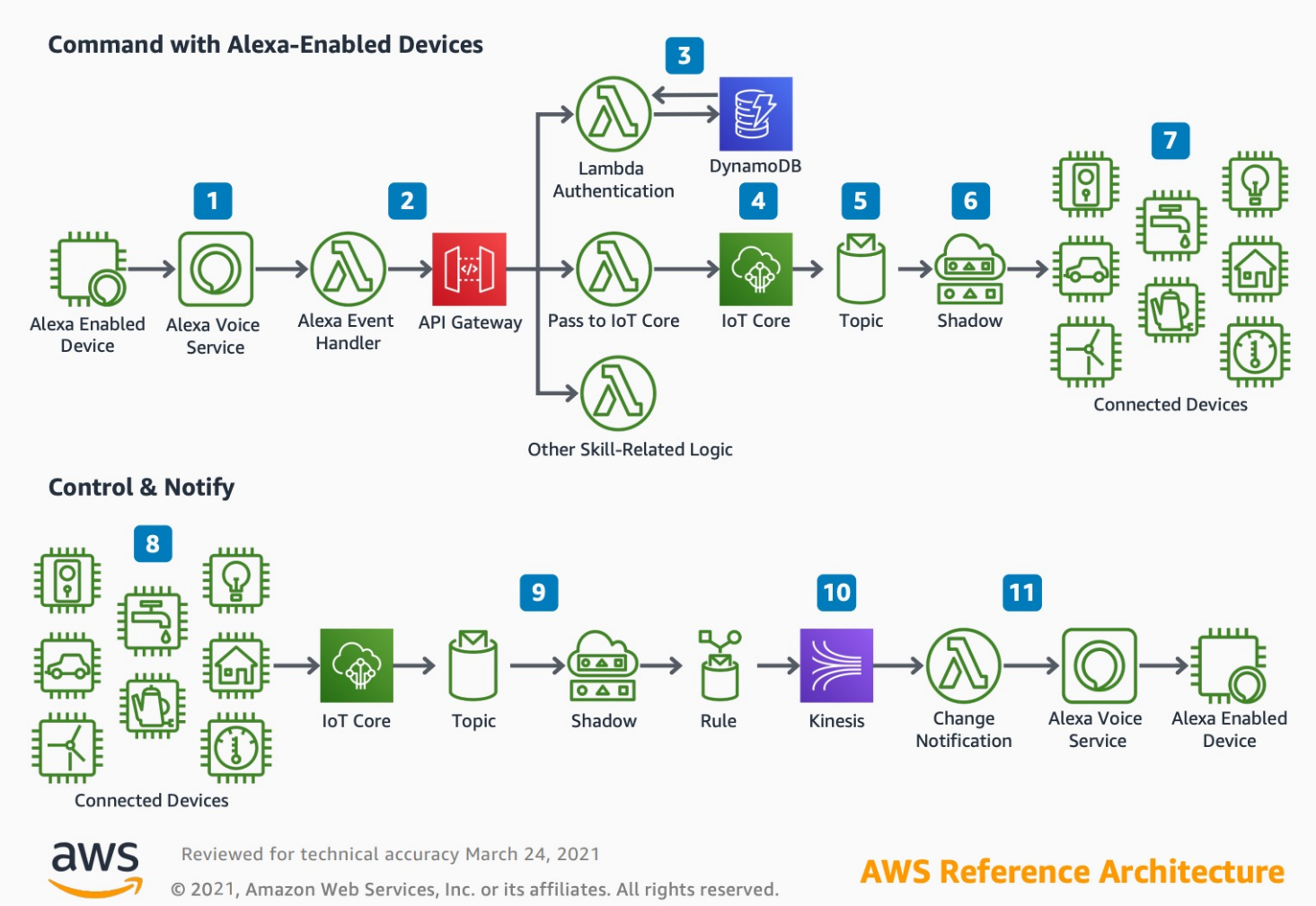
AWS Reference Architecture – Industrial Platform

1. The devices connect to AWS IoT Core, directly via MQTT from FreeRTOS or via AWS IoT Greengrass.
2. An IoT topic Rule can be configured in AWS IoT Core to pick up meaningful data and save it in Amazon DynamoDB.
3. Another IoT topic Rule can be configured in AWS IoT Core to trigger AWS IoT Events for certain data, like temperature etc.
4. Or, directly from AWS IoT Greengrass, data can be ingested in AWS IoT Analytics.
5. Client applications can make an API call to retrieve data via Amazon API Gateway.
6. AWS IoT Events can trigger Amazon SNS actions in case certain changes occur in the data (e.g. Temperature > 25 degrees)
7. Data from AWS IoT Analytics can be imported into Amazon QuickSight, for visualization and further insights.
8. Amazon API Gateway invokes an AWS Lambda function with logic to retrieve data requested by the client applications from Amazon DynamoDB
9. Amazon SNS triggers an email notification for client applications.
10. Client applications can visualize the data in Amazon QuickSight.

Connected Home – Command & Control Integrate Alexa with your connected home devices.



Connected Home – Command & Control Integrate Alexa with your connected home devices.

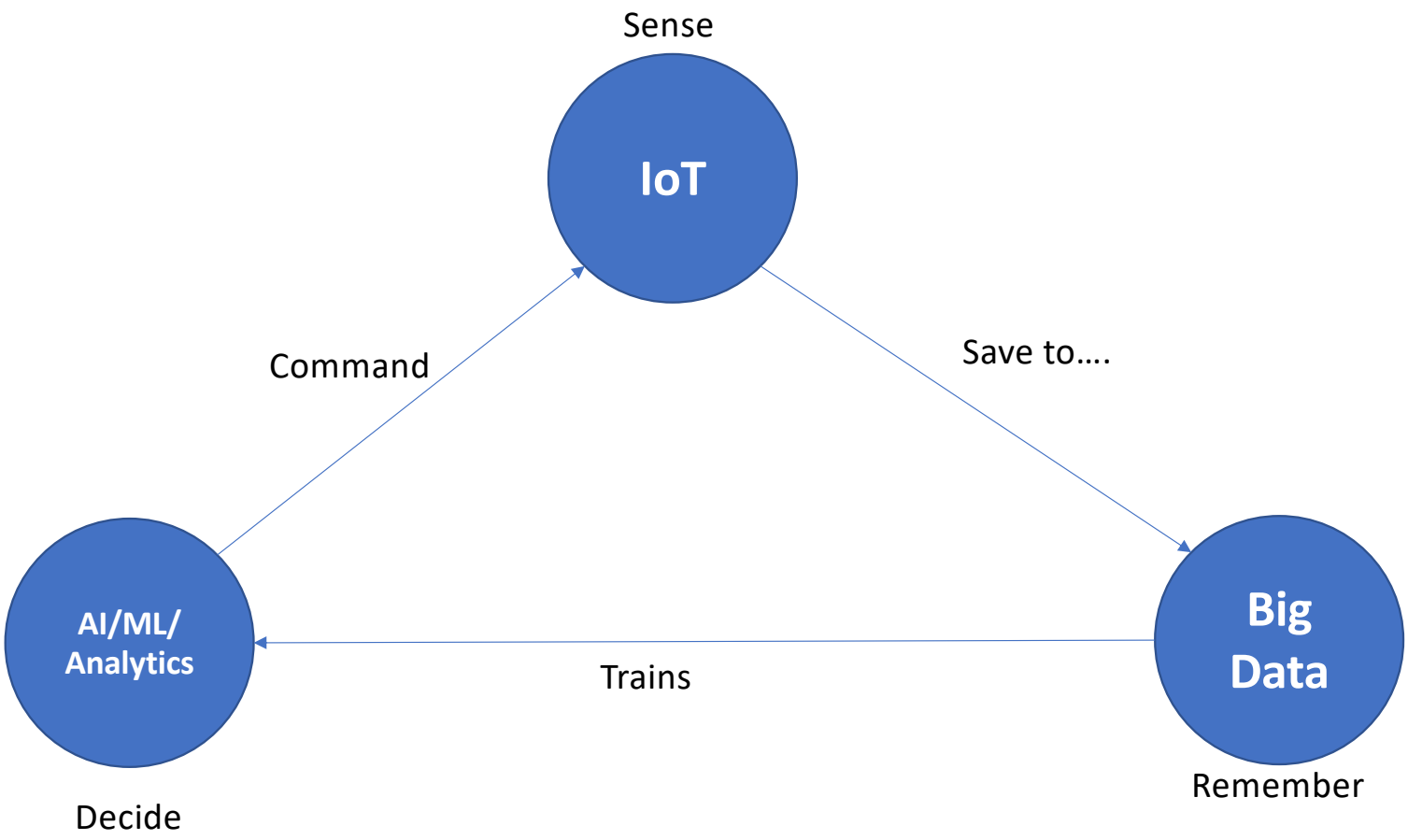
1. An Alexa invocation is created by an Alexa Enabled Device running either the Alexa Voice Services SDK or using an Amazon Echo.
2. Alexa Skill uses AWS Lambda as its backend logic. Amazon API Gateway handles validating the authorization and access control for the Alexa skill before it sends data to the correct Skill Handler.
3. Amazon DynamoDB holds user authorization data that associates your users to their skill and devices while API Gateway verifies whether the request should be accepted.
4. AWS IoT Core enables you to connect devices to the cloud. IoT Core sends and receives updates based on Alexa invocations.
5. Each Alexa invocation triggers an IoT Rule, which evaluates each invocation and routes the message to other AWS Services.
6. Each Alexa invocation triggers a message to the AWS IoT Device Shadow. The Shadow service then sends a message to the device with settings that need to be updated to match the Alexa request.
7. IoT data is sent securely from AWS IoT Core to the devices that need to take action based on the Alexa command.
8. In addition to commands, Alexa also can receive updates of device data that happens locally. IoT data originated by your devices can send data to AWS IoT Core using AWS IoT SDK, AWS IoT Greengrass, or Amazon FreeRTOS.
9. Messages from the devices are sent to IoT Core, which uses an IoT rule to write data to Amazon Kinesis Data Streams.
10. Real-time data is streamed from Amazon Kinesis Data Streams. Amazon Kinesis Data Analytics, Spark (EMR), EC2, Lambda, and other services that extract data for processing.
11. The notifications interface to Alexa Voice Service is invoked in the AWS Lambda function. It gives users a visual or audio indication that new content is available from Alexa Re domains or an enabled Alexa skill.

The pillars of the AWS Well-Architected Framework

Name	Description
Operational Excellence	The ability to support development and run workloads effectively, gain insight into their operations, and to continuously improve supporting processes and procedures to deliver business value.
Security	The security pillar describes how to take advantage of cloud technologies to protect data, systems, and assets in a way that can improve your security posture.
Reliability	The reliability pillar encompasses the ability of a workload to perform its intended function correctly and consistently when it's expected to. This includes the ability to operate and test the workload through its total lifecycle. This paper provides in-depth, best practice guidance for implementing reliable workloads on AWS.
Performance Efficiency	The ability to use computing resources efficiently to meet system requirements, and to maintain that efficiency as demand changes and technologies evolve.
Cost Optimization	The ability to run systems to deliver business value at the lowest price point.
Sustainability	The ability to continually improve sustainability impacts by reducing energy consumption and increasing efficiency across all components of a workload by maximizing the benefits from the provisioned resources and minimizing the total resources required.

https://docs.aws.amazon.com/wellarchitected/latest/framework/welcome.html

IoT and Big Data - Analytics - AI - Machine Learning



IoT and Big Data - Analytics - AI - Machine Learning

- **The IoT creates of massive amounts of data from sensors and now the biggest challenges not only from a transport perspective but also from a data management standpoint.**
- **E.g :**
  - **Commercial Aviation Industry**
  - **Modern jumbo jet is equipped with 10,000 sensors then petabyte (PB) of data generated per day per commercial airplane.**
- **Analyzing this amount of data in the most efficient manner, using data analytics**

Modern jet engines fitted with 1000 of sensors that generate a 10GB of data per second.

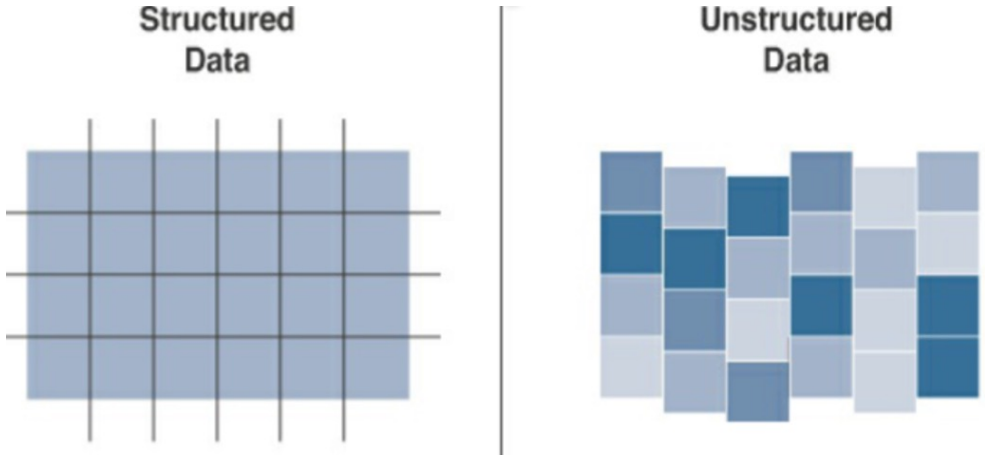
Figure 7-1 Commercial Jet Engine

IoT and Big Data - Analytics - AI - Machine Learning

- **A few key concepts related to data.**
  - **Not all data is the same; most of the time it can be categorized and analyzed**
- **Two important categorizations from an IoT perspective are:**
  - **Data is structured or unstructured**
  - **Data is in motion or at rest.**

- **Structured and unstructured data require different approaches for analysis**

Organized formatting
e.g. spreadsheet, database



Does not conform to a model
e.g. text, images, video, speech

IoT – structured vs unstructured data

- **Structured Data**
  - **Well represented and organized.**
  - **Can be handled by traditional database systems (RDBMS)**
  - **Can be easily formatted, stored, queried and processed**
  - **IoT sensor data uses often structured values, such as temperature, pressure, humidity …. which are all sent in a known format**
- **Unstructured Data**
  - **Does not fit into a predefined data model**
  - **Cannot so easily be handled by RDBMS (NoSQL data come into place)**
  - **Machine learning – cognitive computing etc. can be applied to analyse unstructured data such as**
    - **Natural language processing**
    - **Image recognition**
    - **AI in general**

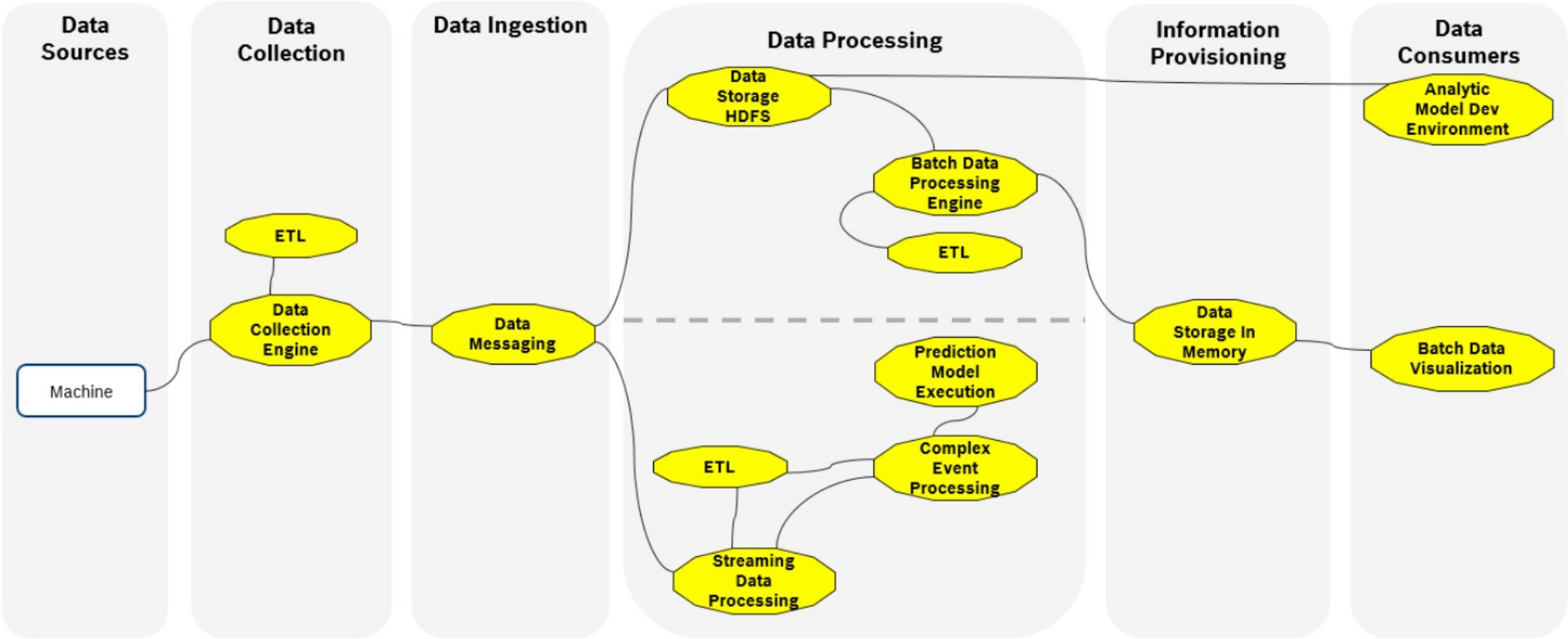
IoT Objects -> smart IoT devices generate structured and unstructured data

IoT – Data in motion vs Data at rest

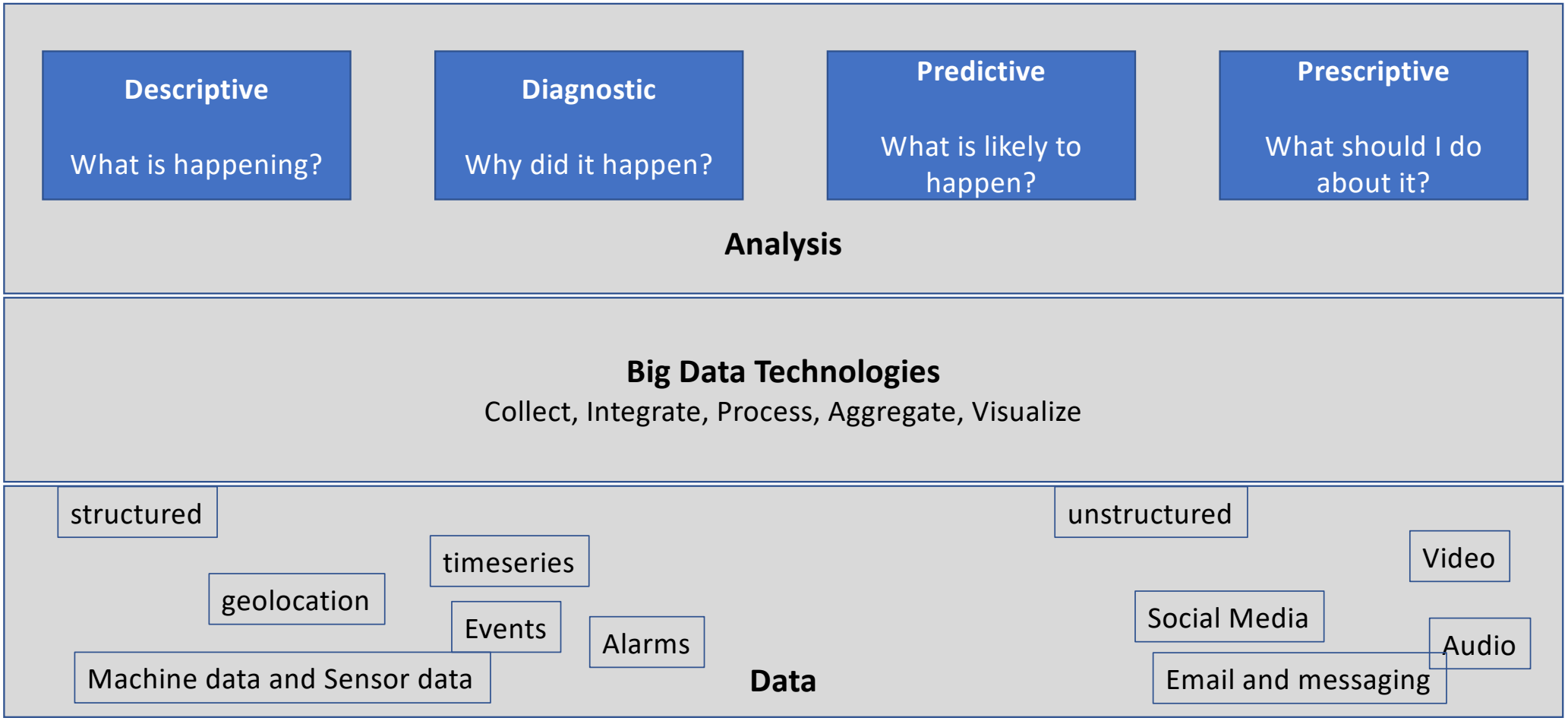
- **Data in motion**
  - **‘data is on transit’**
  - **‘edge processing’ in real-time, filter data, delete data, forward data**
  - **Real time streaming analytics**

- **Data at rest**
  - **‘data held or stored’**
  - **Definition – data that is not actively moving from device to device or network to network** such as data stored on ….
  - **….Data at rest is stored on file servers, databases, documents, flash drives etc.**
  - **Data at rest analysis to get information out of data**

Customer example – IoT Data analytics platform



IoT Data analysis overview

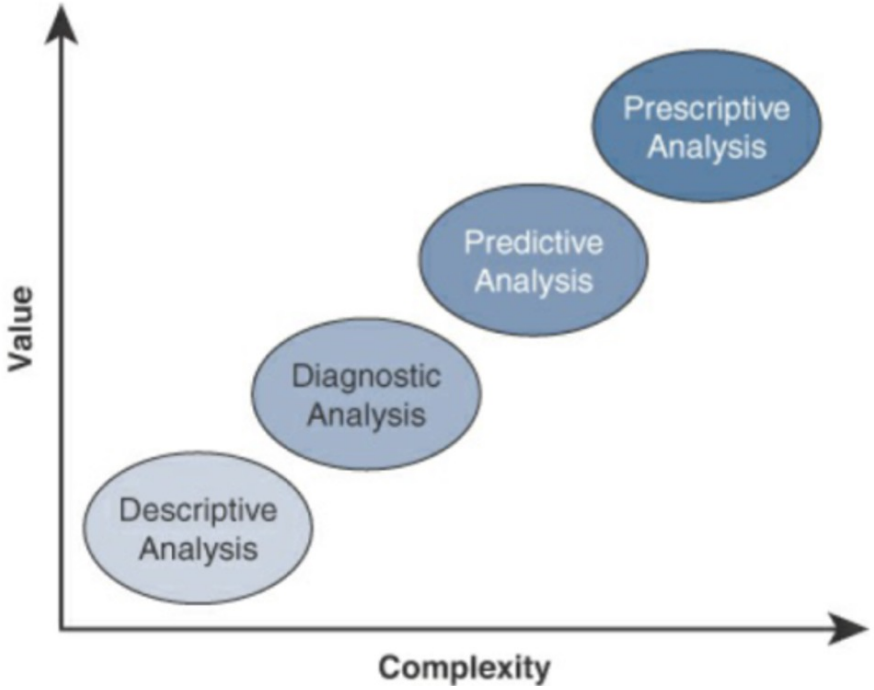


IoT data analytics – small story form descriptive to prescriptive

- **Descriptive**
  - **Descriptive data analysis tells you what is happening, either now or in the past.**
  - A thermometer in a truck engine reports temperature values every second. So we can collect the data at any moment.
  - Too high temperature, indicate problem in a cooling or the engine may be experiencing too much load.
- **Diagnostic**
  - **When you are interested in the "why," diagnostic data analysis can provide the answer.**
  - The temperature sensor in the truck engine, you might get why the truck engine failed.
  - Diagnostic analysis might show that the temperature of the engine was too high, and the engine overheated
- **Predictive**
  - **Predictive analysis aims to foretell problems or issues before they occur.**
  - With historical values of temperatures for the truck engine, predictive analysis could provide an estimate on the remaining life of certain components (Replace parts or oil change or engine cooling maintenance) in the engine.
- **Prescriptive**
  - **Prescriptive analysis goes a step beyond predictive and recommends solutions for upcoming problems.**
  - A prescriptive analysis of the temperature data from a truck engine might calculate various alternatives to cost effectively (frequency of oil changes or upgrading to new engine) maintain truck.

IoT Data analysis overview

Application of Value and Complexity Factors to the Types of Data Analysis



IoT Data analysis challenges

- **There are two specific challenges that IoT places especially to rotational database systems**
  - **Scaling problem**
    - Huge amount of data might be generated
  - **Volatility**
    - RDBMS – it is important and critical to define the correct schema from beginning. Changing afterwards it may result in bad performance and inflexible
- **Therefore NoSQL database are very often used in IoT solutions.**
  - **They support a complex and evolving data model**
    - **NoSQL is not a specific database technologies, has several different types of databases, including**
      - Document stores (e.g. couchDB)
      - Key value stores (e.g. MongoDB)
      - Wide column stores (e.g Apache Casandra)
      - Graph stores (e.g. Neo4j)

IoT – Big data analytics tools and technology

- **Terms big data and Hadoop interchangeably**
  - Hadoop is at the core of today's big data implementations
  - Big data analytics can consist of many different software pieces that together collect, store, manipulate, and analyze all different data types.

- **"Three Vs" to categorize big data**
  - Velocity: Velocity refers to how quickly data is being collected and analyzed.
    - E.g. Hadoop Distributed
  - **Variety: Variety refers to different types of data. Often you see data categorized as structured, semi-structured, or unstructured.**
    - E.g. Hadoop is collect and store all three types.
  - **Volume: Volume refers to the scale of the data (GB to PB to EB (Exa))**
    - E.g. Big data

What is Apache Hadoop?

Apache Hadoop is an open-source software utility that allows users to manage big data sets (from gigabytes to petabytes) by enabling a network of computers (or “nodes”) to solve vast and intricate data problems.

It is a highly scalable, cost-effective solution that stores and processes structured, semi-structured and unstructured data (e.g., Internet clickstream records, web server logs, IoT sensor data, etc.).

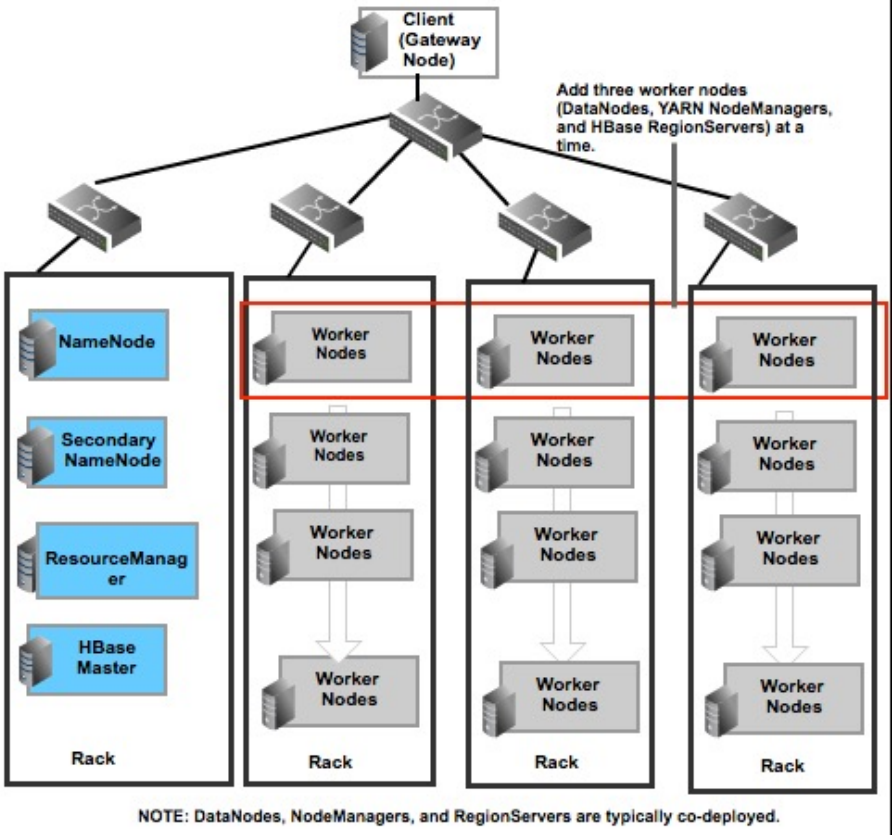
Benefits of the Hadoop framework include the following:

- Data protection amid a hardware failure
- Vast scalability from a single server to thousands of machines
- Real-time analytics for historical analyses and decision-making processes

- Hadoop consist of
  - HDFS File System (Hadoop distributed file system to store data across multiple nodes and
  - MapReduce a distributed processing engine that splits large task into smaller once which can run in parallel

https://www.ibm.com/cloud/blog/hadoop-vs-spark

Hadoop - typical cluster architecture

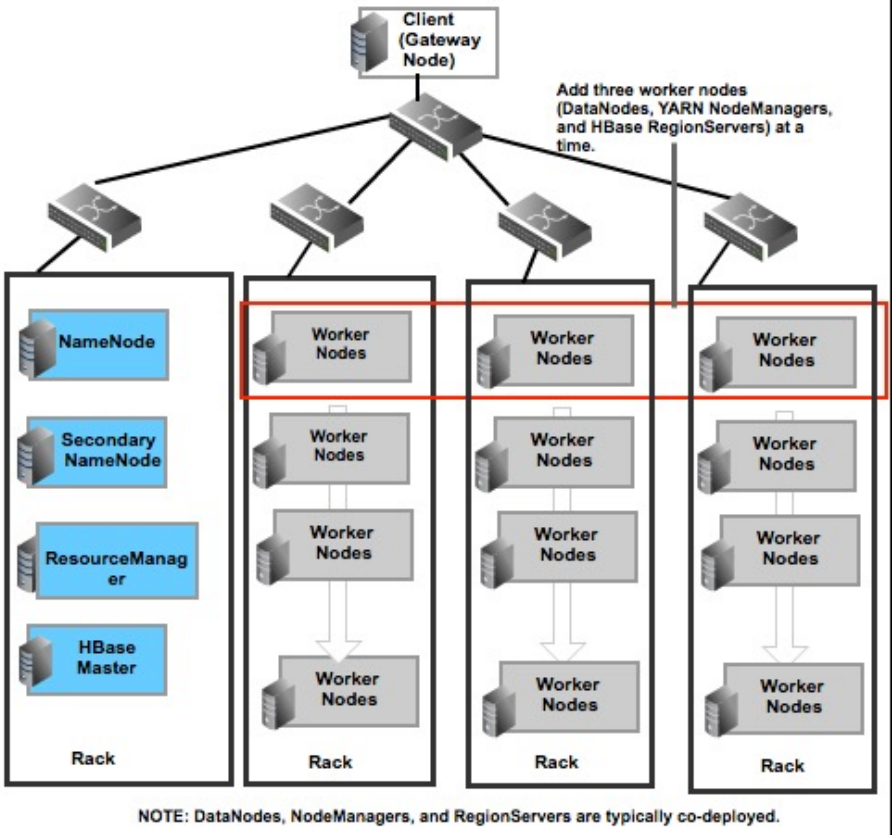


HDFS – the capability is handled by specialized nodes in the cluster
The NameNodes and DataNodes

The NameNode takes write requests from clients and distributes thouse files across available nodes in configurable block sizes, usually 64MByte or 128MByte blocks

DataNodes are the servers where the data is stored
Data blocks are distributed across several nodes and often they are replicated for redundancy.

Hadoop - typical cluster architecture

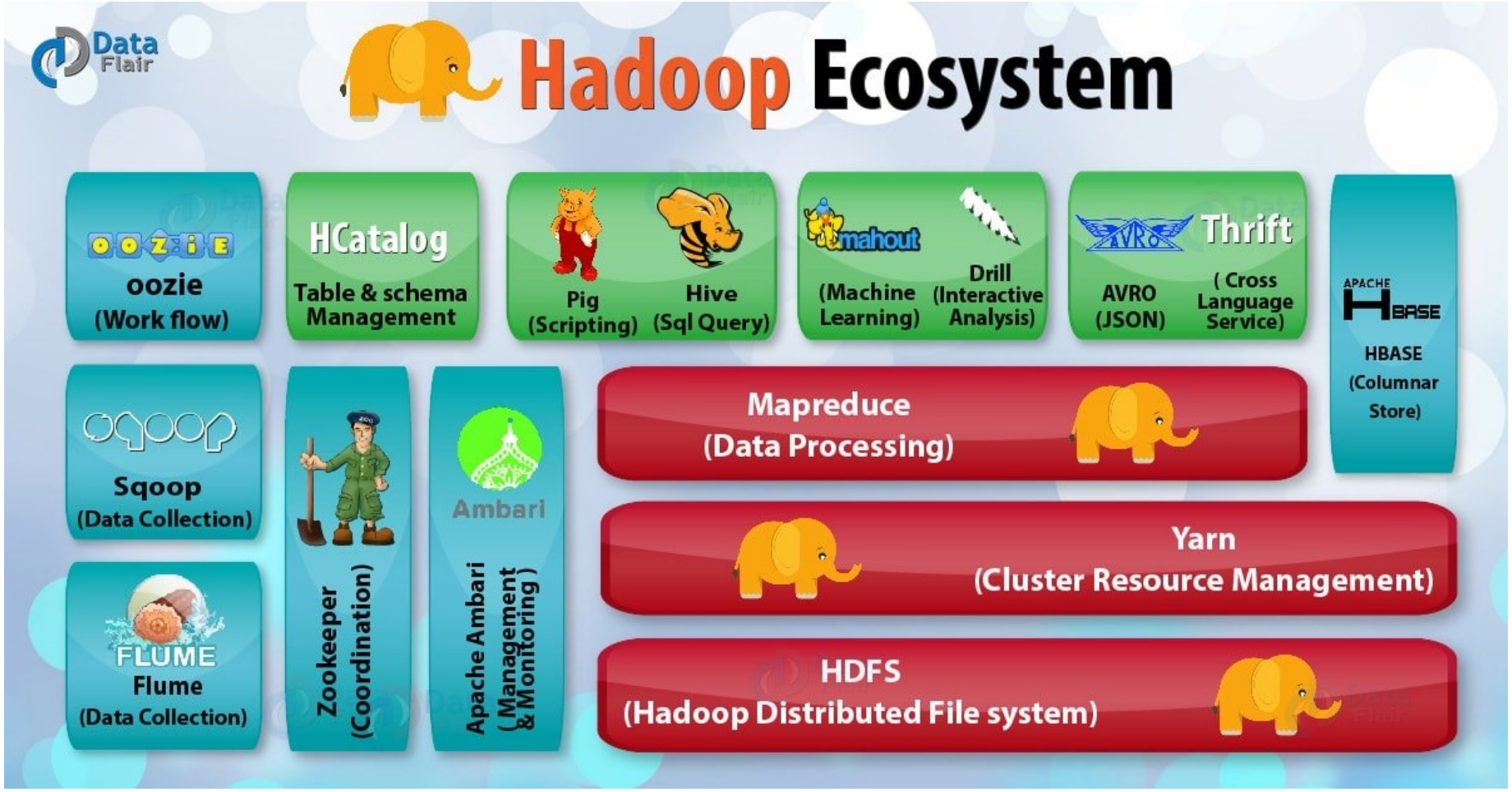


Hadoop batch processing
Running a scheduled or ad hoc query across historical data stored in the HDFS

Query is broken into smaller tasks and distributed across all the nodes running MapReduce in a cluster.

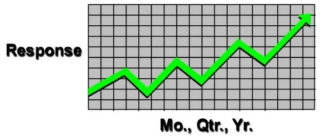
Attention: Not useful for real time processing

Today Hadoop is more than HDFS and MapReduce - a complete Ecosystem

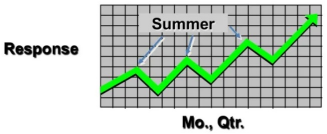


IoT Time series data and some challenges in analysing the data

- **Trend detection**



- **Seasonal component**



- **Irregular / Random / Noise Component**

- **Some obstacles**
  - **Duplicate values, missing values, wrong values, out of range values, ……**

  …..

IoT Time series data and some challenges in analysing the data

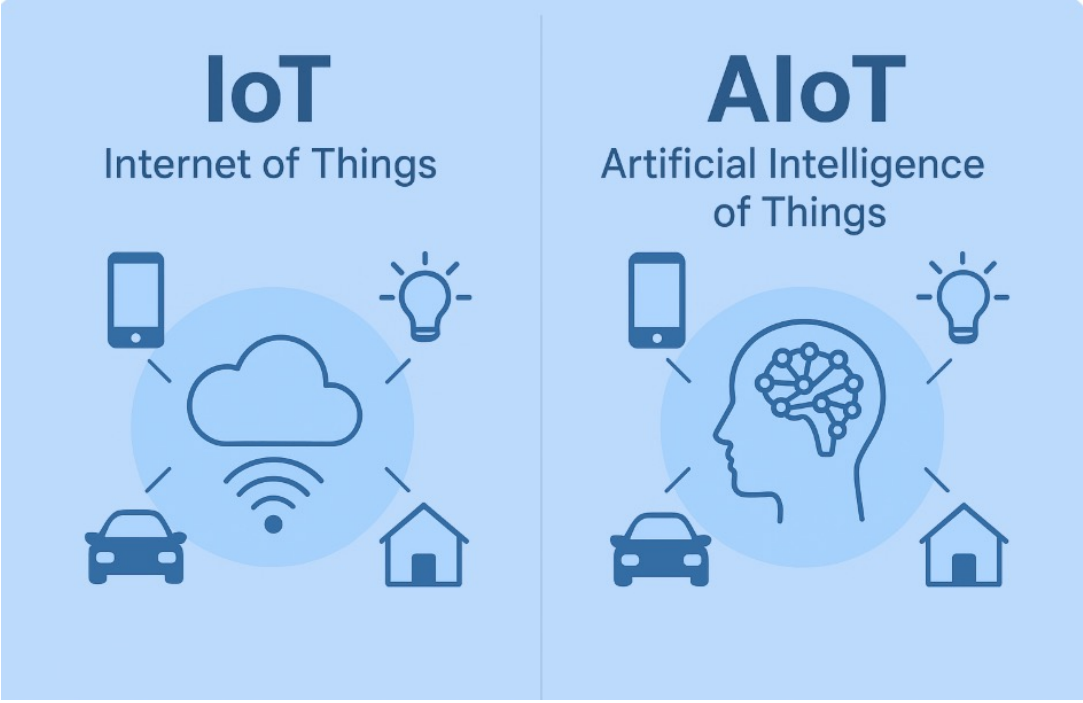
Let’s have a short look to Kaggle

a big AI & ML Community

kaggle to Seasonality

- What is Seasonality?
  - We say that a time series exhibits **seasonality** whenever there is a regular, periodic change in the mean of the series. Seasonal changes generally follow the clock and calendar -- repetitions over a day, a week, or a year are common. Seasonality is often driven by the cycles of the natural world over days and years or by conventions of social behavior surrounding dates and times.
-

IoT and AIoT (Artificial Intelligence of Things)



Source: Diagram generated by chatgpt

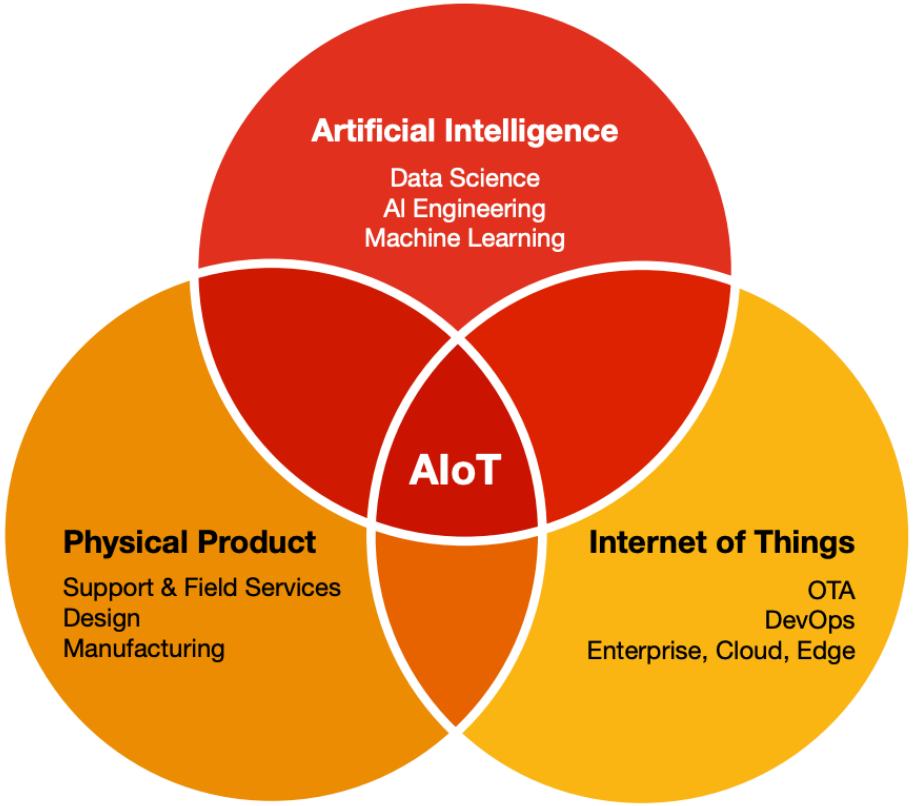
AIoT (Artificial Intelligence of Things) combines IoT devices with AI capabilities, typically bringing analytics, automation, and decision-making **closer to the edge**.

We already discussed it without explicitly name it AIoT

AIoT

- **AIoT is the integration of IoT and AI to achieve**
  - **Intelligent decision making**
  - **Real-time automation**
  - **Predictive and adaptive behaviour**
- **IoT provides the data of the connected devices**
  - **In IoT we collect the data, store it and analyze the data in the data store**
  - **In AIoT we extract the insights and make autonomous decisions.**
  - **AIoT is used in smart city, smart home, smart factory, and in your smart solution (?)**

AIoT as a combination of a physical product, IoT and AI



Source: pwc-whitepaper-trusted-aiot.pdf

The evolution to AIoT

Siloed IoT
System core controls the devices

Sensing and Monitoring
- Collection of environment data
- Monitor environment related issues

Connected IoT
Devices are interconnected to interact together

Manage and Control
- Unification, cleaning and classification of data gathered via sensors

Augmentated IoT
Connected devices are merged with AI to enhance the value

Augmentation
- Analyzing of data from sensors to generated insights

AIoT
Intelligence driven sytems of connected deviceses are build to carry out autonomous operations

Autonomy
- Use of AI to decide the best possible action over multiple connected devices

AIoT Solution – Industrial Equipment Anomaly Detection at the Edge

1 IoT Sensors

- Attach vibration, temperature, and current sensors to motors, pumps, or turbines
- Sensors continuously collect streaming data

2. Edge Device with AI Model

- Lightweight AI/ML model deployed on an edge device (e.g., NVIDIA Jetson Nano, Raspberry Pi + Coral USB Accelerator)
- Model is trained to learn normal operating patterns of the machine
- Detects anomalies in real-time, e.g., unusual vibration or temperature spikes

3. Real-Time Alerts and Actions

- Edge device flags anomalies instantly without cloud communication
- Sends alerts to operators via dashboard, SMS, or automated system
- Can trigger automatic safety shutdown or reduce machine load to prevent damage

4. Cloud AI for Model Improvement

- Aggregate anomaly data over multiple machines
- Retrain models to improve accuracy and detect new failure patterns

Algorithm used in AIoT

Algorithm Type	Examples	AIoT Use Case
Machine Learning	Random Forest, SVM	Predictive maintenance
Deep Learning	CNN, LSTM, Autoencoders	Smart cameras, forecasting
Reinforcement Learning	Q-Learning, DQN	Traffic/energy optimization
TinyML	Quantized neural nets	Edge devices with low power
Anomaly Detection	Isolation Forest	Security & equipment monitoring
Computer Vision	YOLO, U-Net	Smart surveillance
Time-Series Forecasting	ARIMA, Transformers	IoT sensor prediction
Federated Learning	Distributed models	Privacy-friendly edge AI

Keep in mind: AIoT needs lightweight models that run on low-power devices.

China has the most 5G stations deployed – here some showcases on 5G technology

Smart street lights: an important entrance for smart city construction

5G Smart Cities:

Cities like Shenzhen and Beijing have implemented 5G networks to support smart city initiatives, including traffic management, public safety, and smart lighting systems that optimize energy use.

China Telecom's 5G Helped Beijing Jishuitan Hospital Perform the World's First Remote Robotic Surgery

Telemedicine:

Hospitals, particularly in rural areas, have used 5G to provide telemedicine services, allowing doctors to perform remote consultations and even remote surgeries through robotic systems, ensuring better healthcare access.

Autonomous Vehicles:

Chinese companies like Baidu and Huawei have been testing autonomous driving technologies with 5G connectivity, enhancing vehicle-to-everything (V2X) communication to improve safety and efficiency.

Agriculture:

5G technology is being applied in smart agriculture for precision farming. Farmers use drones and IoT devices connected through 5G networks to monitor crop health, automate irrigation, and optimize resource use.

Education:

Remote learning programs have adopted 5G to support interactive learning experiences, including virtual classrooms that enable high-quality video communication and collaborative tools.

Retail:

Retailers are integrating 5G to enhance customer experiences through augmented reality applications, smart fitting rooms, and personalized marketing.